

**Third Semester B. Tech. (Electronics and Communication Engineering)
Examination**

NETWORK THEORY

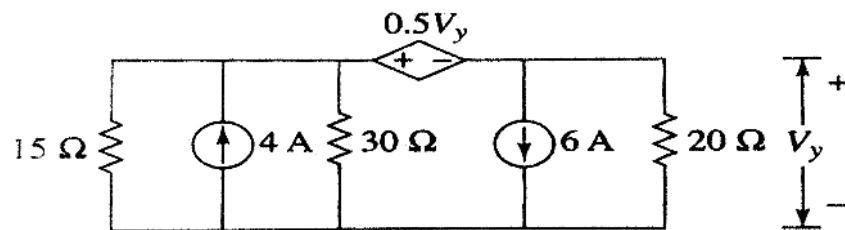
Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

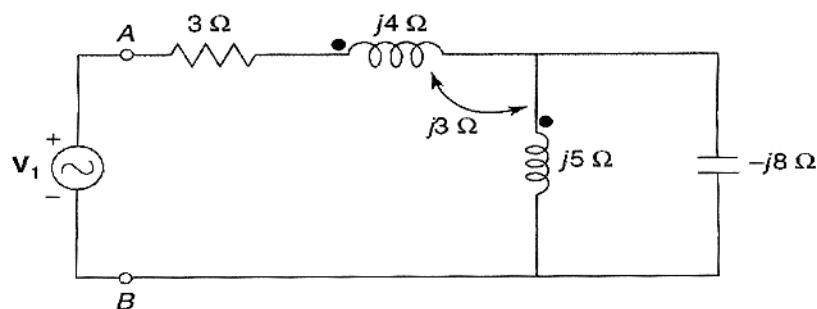
- (1) Draw neat and clean diagram, wherever necessary.
- (2) Use of non-programmable calculator is permitted.
- (3) Assume suitable data wherever necessary.
- (4) Solve any two questions out of Q. 6 (a), (b), and (c).

1. (a) Evaluate the following network containing dependant sources and determine all independent nodal voltages using nodal analysis. (Down line is the reference line.)



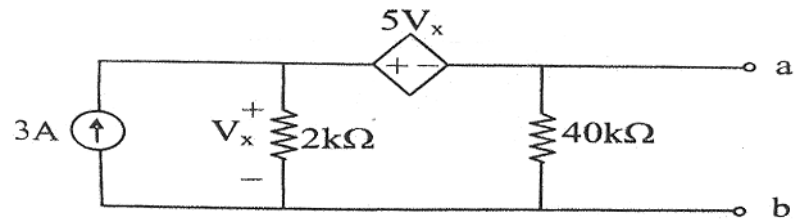
5(CO1)

- (b) In the below given coupled circuit, estimate the value of input impedance across terminals A and B.



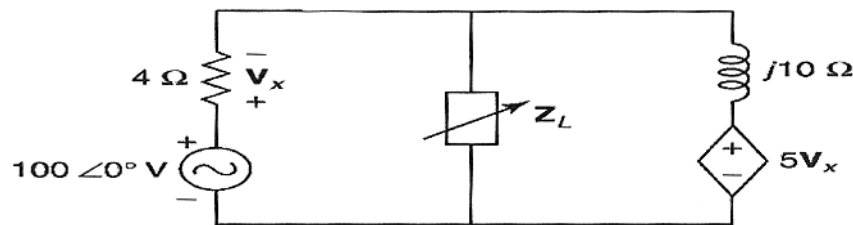
5(CO1,3)

2. (a) Estimate the Norton's equivalent circuit across the terminals a and b in below given network.



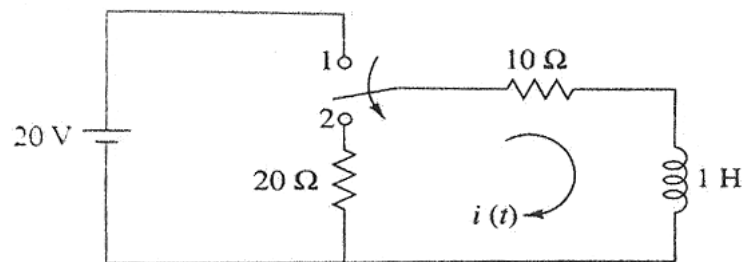
5(CO1,3)

- (b) For the network shown below, evaluate the value of Z_L that will transfer maximum power from the source. Also determine maximum power delivered to load impedance Z_L .



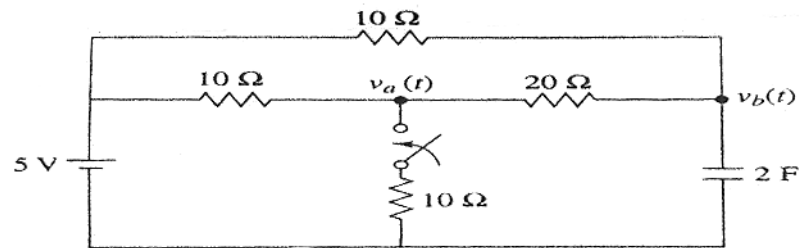
5(CO1,3)

3. (a) In the network shown below a steady state is reached with s/w is at position 1. At $t = 0$ s/w is moved to position 2, find the current $i(t)$ in the network at $t > 0$ sec, using Laplace transform.



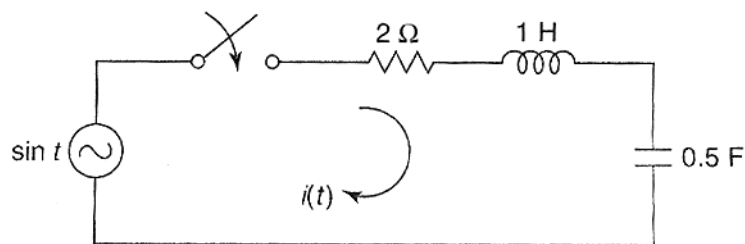
5(CO1,2)

- (b) In the network shown below a steady state is reached with switch is open. At time $t = 0$, s/w is closed. Determine nodal voltage $v_a(0^-)$, $v_b(0^-)$, $v_a(0^+)$, $v_b(0^+)$.



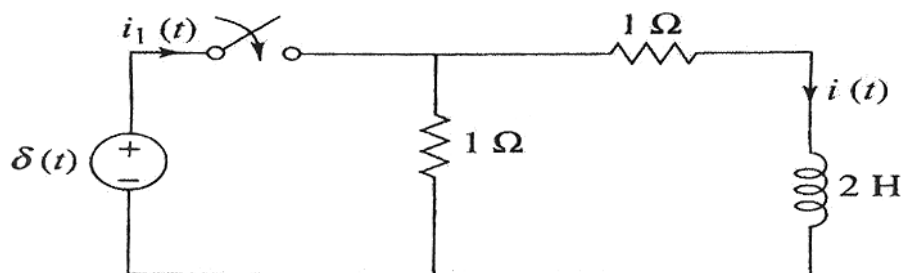
5(CO1,2)

4. (a) Evaluate the network shown below after the switch is closed at $t = 0$. Determine the current $i(t)$ for $t > 0$, assuming zero initial conditions. (Switch is closed at $t = 0$ with all initial conditions zero.)



5(CO1,2)

- (b) Determine the current $i(t)$ at $t > 0$ in the network shown below using Laplace and inverse domain. (Switch is closed at $t = 0$ with all initial conditions zero.)

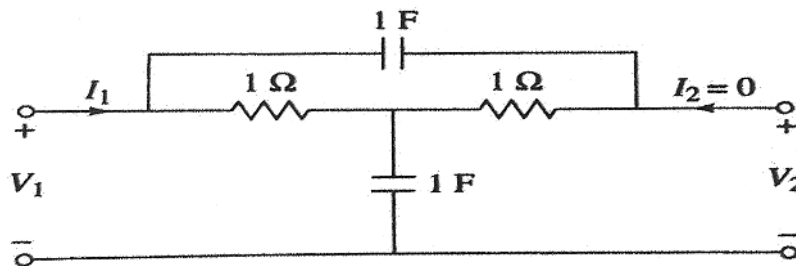


5(CO1,2)

5. (a) Given the following network function $F(s)$ in Laplace domain :

$$F(s) = \frac{10 (s^2 + 9) (s - 2)}{(s^2 + 1) (s^2 + 2s + 5)}$$

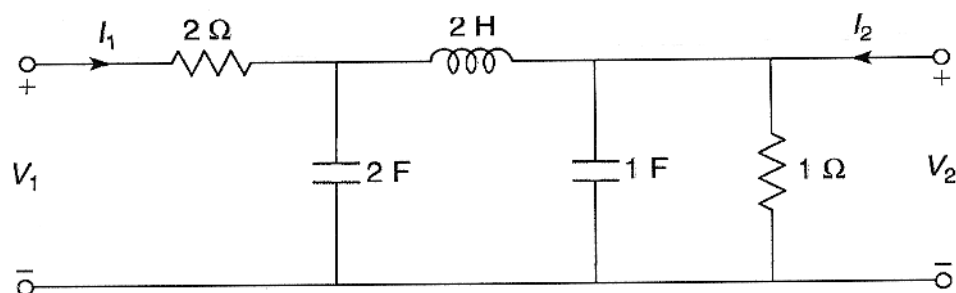
- (i) Plot the pole zero locations of $F(s)$ in s -plane.
 - (ii) Is it Driving point function or Transfer function ? Justify your answer.
 - (iii) Find the DC gain (scaling factor) of $F(s)$.
 - (iv) When will this network function becomes **unstable** based on the location of poles in s -plane ? 5(CO4)
- (b) For the network shown below estimate the impedance function $Z_{11}(s)$ and voltage transfer function $G_{12}(s)$.



5(CO1,4,5)

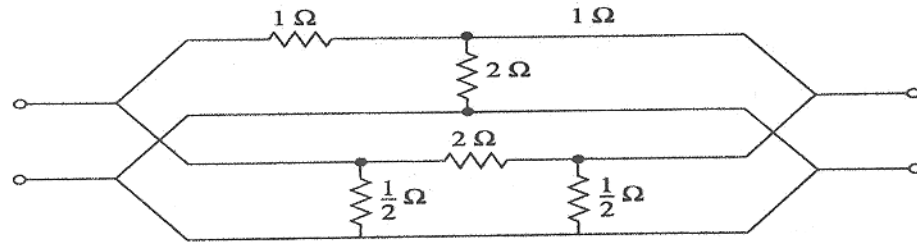
6. Solve any **Two** questions out of Que. 6 (a), (b) and (c) :

- (a) Evaluate the ABCD parameters for the network shown below. Also check for the symmetry and reciprocity property in this network. (In S-domain)



5(CO1,5)

- (b) Evaluate the Y-parameters of the below given 2-port network using interconnection of networks. Also check for the symmetry and reciprocity property in this network.



5(CO5)

- (c) Design a T-section constant-k high-pass filter having a cut-off frequency of 10 kHz and a design impedance of 600 Ω . Find its characteristic impedance and phase constant at 25 kHz.

5(CO4,5)

